

# Introductory Data Science Labs

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# Abstract

Introductory  
Data Science  
Labs

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Future Work

*Lab materials provide students opportunities for practice in new domains. However, existing tools for implementing interactive data science labs require installing software, which distracts from core lessons. We leverage R's instructional ecosystem for browser-based, self-guided Python tutorials for data science courses. These tutorials require minimal setup, ideal for use across different classroom modalities. R's framework can be successfully adapted for Python labs, lowering barriers to entry and strengthening student learning of core data science concepts.*

# Data Science as an Emerging Field

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The field of data science has rapidly expanded in recent years, and approaches to *designing introductory data science course materials* vary widely, depending on the background of the instructor (Hardin et al., 2015).

- Statistics departments: R and its extensive range of pedagogical tools
- Computer Science departments: Python
- Still others: language agnostic

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# Tools for Designing Course Materials

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- Python-based courses rely on Jupyter for interactive work.
- R's more mature teaching libraries do not run in Jupyter.
- Comparable Python tools either:
  - Do not work as well, or
  - Are unfamiliar to statistics faculty new to teaching Python (Brunner & Kim, 2016).

Answer: Shiny Apps, written in R to teach Python, combined with PositCloud

- Simple, interactive assignments,
- Lowering barriers for beginners (Çetinkaya-Rundel and Rundel, 2018; Stoudt et al., 2022).

# Motivation

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Consider teaching a group of students how to create a histogram:

- Show students what a Jupyter notebook is and how it works (i.e., the difference between code and markdown cells, shortcuts, etc.).
  - Open a Jupyter notebook. Either:
    - Open a terminal and launch an empty notebook.
      - Show students what the terminal is.
      - Show students how to change their directories.
    - Download and install Anaconda Navigator.
  - Install relevant graphics modules.
    - Show students what the terminal is.
    - Show students how to install packages.
  - Load data into your workspace.
    - Download the dataset.
    - Save the dataset in the same directory as the notebook.
      - Review the file structure of your computer to understand how to navigate to a new directory.
  - **Actually make the histogram.**
- Click on a link.
  - Sign into the website.
  - **Actually make the histogram.**

# Example 1: Histogram

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8. Do you see a warning message saying "Pick better value with 'binwidth'"? This error is unique to `geom_histogram()`. To fix the warning, you can change the number of bins by adding a new argument, `bins = 10`, into the `geom_histogram()` function.

Python Code

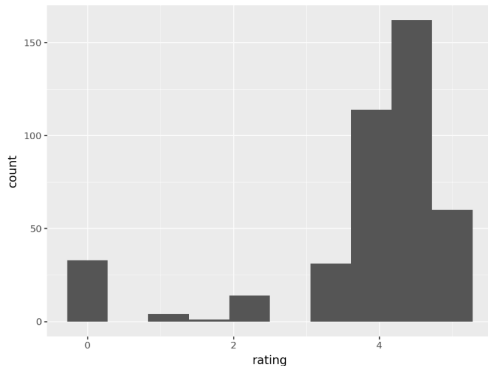
Start Over

Solution

Run Code

Submit Answer

```
1 (p9.ggplot(epicurious_num_nd, p9.aes(x = 'rating'))  
2 + p9.geom_histogram(bins = 10)).show()
```



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## Example 2: Question Types

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5. It is also helpful to discuss how many missing values the dataset has. Identify the percentage of missing values in the `protein` column.

Python Code

 Start Over

 Solution

 Run Code

```
1 |  
2 |  
3 |
```

What percent of `protein` is missing? (Round to 1 decimal place.)

Submit Answer

6. Is it appropriate to drop the missing data?

☐ Yes

☐ No

Submit Answer

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# Example 3: Responses

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```
Python Code ↺ Start Over 🔍 Solution ▶ Run Code  
1 sum(epicurious_num['protein'].isna())/len(epicurious_num['protein'])  
2  
3
```

0.18482490272373542

What percent of **protein** is missing? (Round to 1 decimal place.)

20

Incorrect

Try Again

6. Is it appropriate to drop the missing data?

☒ Yes

☒ No

Correct!

Congratulations! You have found the second secret word: SAY.

# Research Questions

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- **RQ1:** Can frameworks developed for R-based courses be adapted for teaching data science in Python?
- **RQ2:** Do browser-based, interactive labs improve understanding of data science concepts?
- **RQ3:** Do such labs reduce barriers for students from non-coding backgrounds and increase confidence and self-efficacy?

# RQ1

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Can frameworks developed for R-based courses be adapted for teaching data science in Python?

*Yes! With some caveats—but mostly yes.*

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# Statistics by Class and Section

| Quarter      | Class    | Section   | Students    | Format                    |
|--------------|----------|-----------|-------------|---------------------------|
| Winter 2025  | 119      | 02        | 34          | Required In-Class         |
|              | 119      | 03        | 33          | Asynchronous              |
| Spring 2025  | 119      | 01        | 22          | Asynchronous              |
|              | 119      | 02        | 39          | Required, In-Class        |
| Sept. 2025   | 119      | 01        | 12          | Asynchronous              |
| Autumn 2025  | 119      | 01        | 38          | Asynchronous/Optional Lab |
| Winter 2026  | 119      | 01        | 19          | Required Lab              |
|              | 119      | 02        | 44          | Required Lab              |
|              | 120      | 01        | 29          | Various                   |
| Spring 2026  | 119      | 01        | 41          | Required Lab              |
|              | 119      | 02        | 37          | Required Lab              |
|              | 120      | 01        | 25          | Various                   |
| <b>Total</b> | <b>2</b> | <b>12</b> | <b>373*</b> |                           |

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# DATA119 Tutorials

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| Lab     | Sessions | Median Time (m) | Topics                                                                                   |
|---------|----------|-----------------|------------------------------------------------------------------------------------------|
| Lab 1   | 1155     | 39              | Data wrangling, summary statistics, data visualization                                   |
| Lab 2   | 1557     | 33              | More data visualization, correlation, simple linear regression, some assumption checking |
| Lab 3   | 1248     | 34              | More assumption checking, multiple linear regression, checking significance              |
| Lab 3.5 | 1059     | 34              | Categorical variables, multicollinearity, variance inflation factor                      |
| Lab 4   | 1048     | 40              | Logistic regression, including new model fit statistics like the confusion matrix        |
| Lab 5   | 1587     | 47              | Cross validation, ridge regression, LASSO                                                |
| Lab 6   | 839      | 36              | $k$ -nearest neighbors classification and regression                                     |
| Lab 7   | 1182     | 26              | $K$ -means and hierarchical clustering                                                   |
| Lab 8   | 931      | 41              | Structured Query Language, including joins                                               |
| Lab 9   | 482      | 48              | DATA119 Quarter Review                                                                   |

# DATA119: Usage by Time

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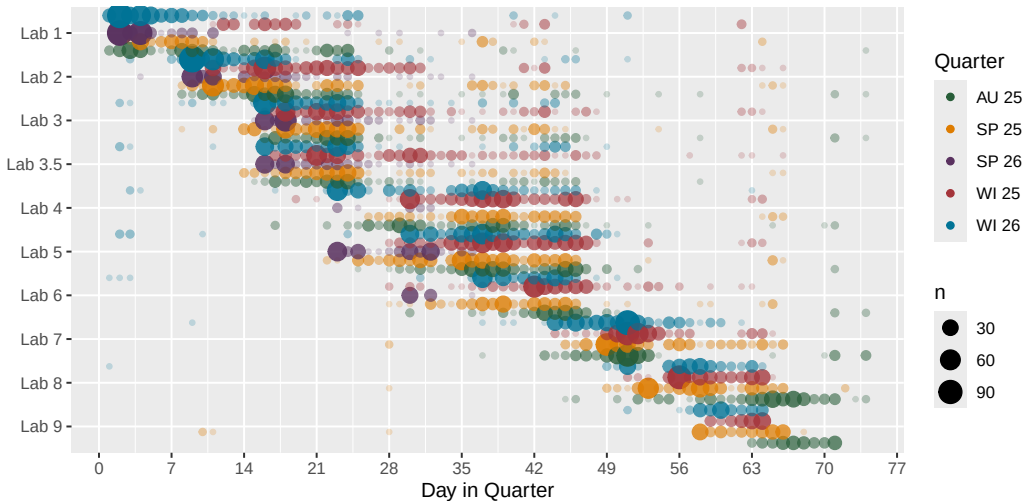
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# Usage for Lab 1: Data Wrangling

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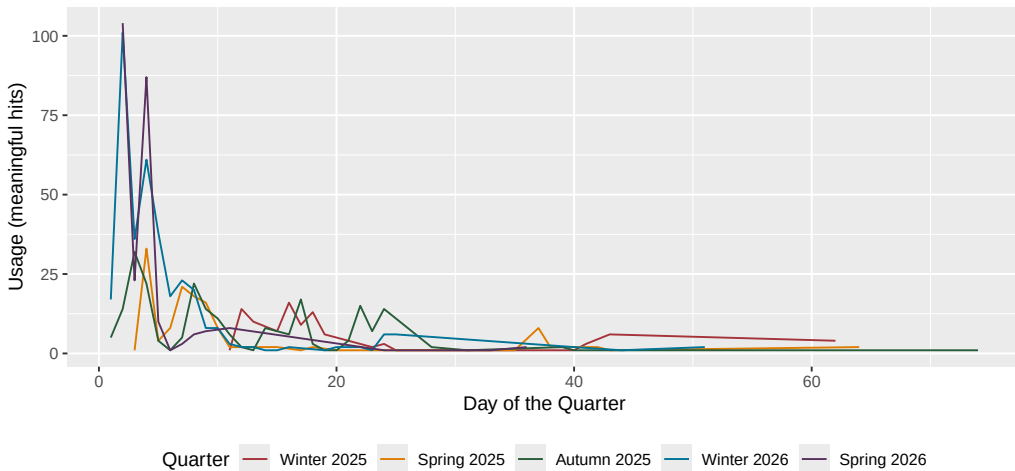
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# DATA120 Tutorials

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| Lab                    | Sessions | Median Time (m) | Topics                                                                                      |
|------------------------|----------|-----------------|---------------------------------------------------------------------------------------------|
| Documentation*         | 18       | 34              | Reading and writing documentation for Python (review)                                       |
| Debugging*             | 54       | 38              | Basic error types and debugging Python scripts (review)                                     |
| Command Line Interface | 201      | 43              | Using the command line interface to navigate through computers and run commands and scripts |
| Style Guide            | 212      | 32              | Motivating and adhering to a style guide for code                                           |
| String Parsing         | 164      | 35              | String logic and methods with relevant applications                                         |
| Sequences of Data      | 192      | 30              | Lists, tuples, dictionaries, and sets in Python                                             |
| Writing Functions      | 24       | 29              | Functions logic and keywords, raising error messages                                        |
| Trees                  | 155      | 50              | Trees as a data structure, some recursive functions                                         |
| Regular Expressions    | 172      | 23              | Regular expressions logic and operators, Python commands in the re module                   |

# DATA120: Usage by Time

10/27

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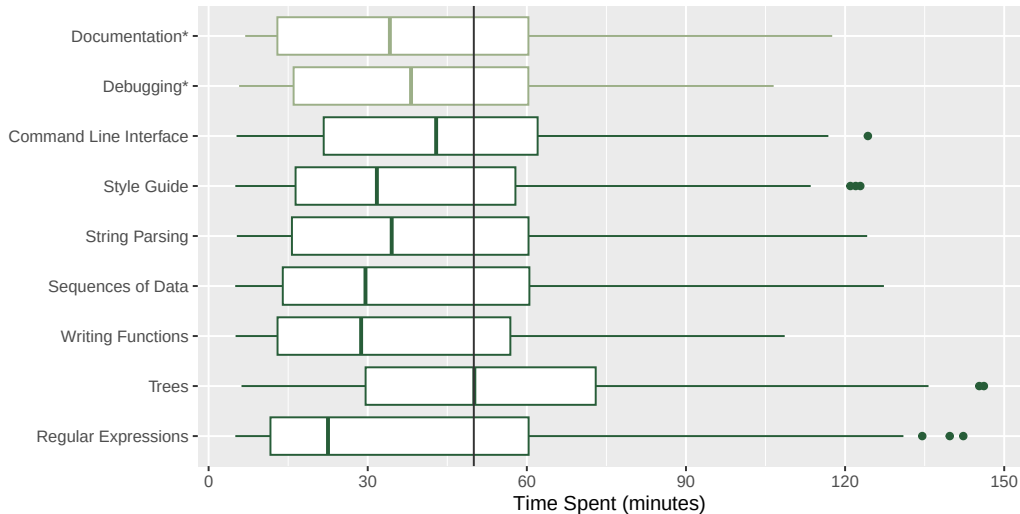
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Summary



# RQ2

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Do browser-based, interactive labs improve understanding of data science concepts?

*Tentatively yes—we need more data.*

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# Anecdata

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In courses without required lab sections, we noticed:

- More student questions that could have been answered by reviewing tutorials
- Delayed understanding, leading to a slightly slower pace in the course, and
- Decreased server performance when students were all trying to complete the lab extremely quickly before the deadline.

# Winter 2026 Statistics

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- **How successful do you think you were in Data 119?**
  - 1: Neutral
  - 7: Mostly Successful
  - 3: Very Successful
  
- **How would your answer to the previous question have changed without the DATA119 labs?**
  - 2: My answer would have changed a little bit, I would have felt more successful.
  - 4: My answer would not have changed at all.
  - 4: My answer would have changed a little bit, I would have felt less successful.
  - 1: My answer would have changed a lot, I would have felt much less successful.

# RQ3

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Do such labs reduce barriers for students from non-coding backgrounds and increase confidence and self-efficacy?

*Not enough data just yet—we will need to continue collection.*

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# Lessons Learned 1

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Server performance matters!

*Sometimes the website crashes, and there is nothing we can do about it, but we still have to stay until the end of the session.*

*The biggest challenge was the one or two weeks when it lagged and we were not allowed to leave and couldn't even work on it.*

Fixes and Future Plans:

- Individual fine-tuning on each tutorial to improve performance
- Reconfiguring server settings to clear up memory
- Migration to a new, larger, Data Science Institute-owned server over the summer.

# Lessons Learned 2

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- Recruitment is hard!
  - Two sections of DATA119 in Winter 2026 with 44 and 19 students
  - Ultimately, only 11 participants completed the pre- and post-quarter surveys.

## Fixes and Future Plans

- Repeated in-class visits by study recruiter
- Canvas announcements with links

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# Future plans

- Adjusting all labs to grade on results rather than exact match code.
- Adjusting grading (currently completion based, but relatively easy to breeze through)
- Ongoing data collection for DATA119 Labs
- Factor Analysis for validating constructs in pre- and post-quarter surveys
- Revision and improvements to DATA120 Labs
- Creation of DATA118 Labs

# Acknowledgments

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